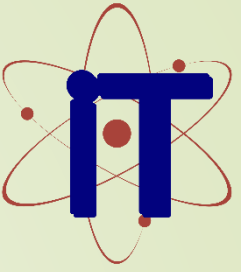




ACROPOLIS
Enlightening wisdom



Presentation
on

Business Analytics Dashboard and Sales Forecasting

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1. Abstract

Business Analytics Dashboard

- ❑ A web-based analytics platform designed to transform raw business data into meaningful insights.
- ❑ Integrates Sales Management, Inventory Monitoring, Reporting, and AI-Based Forecasting into a single system.
- ❑ Enables businesses to track KPIs, monitor stock levels, analyse sales performance, and forecast future demand.
- ❑ Built using the MERN Stack with Machine Learning models for predictive analytics.
- ❑ Helps organisations make faster, data-driven decisions through interactive dashboards and automated insights.

2. Introduction

What is a Business Analytics Dashboard?

- A centralised platform for monitoring and analysing business operations.
- Collects sales and inventory data from various sources.
- Processes and visualises information through interactive charts and reports.
- Provides predictive insights using forecasting algorithms.
- Supports decision-makers with real-time business intelligence and AI-powered recommendations.



3. Problem Statement

Existing Challenges

- ❑ Manual report generation is time-consuming and error-prone.
- ❑ Business data is scattered across multiple systems.
- ❑ Traditional tools focus only on historical analysis.
- ❑ Lack of predictive capabilities for future planning.
- ❑ Poor visibility of inventory levels and stock movement.
- ❑ Delayed business decisions due to the absence of real-time analytics.
- ❑ Enterprise BI tools are often expensive and difficult to customise.



4. Proposed Solution

Business Analytics Dashboard

- ❑ Centralised analytics platform for business monitoring.
- ❑ Dataset upload and automated data processing.
- ❑ Sales analytics and revenue tracking.
- ❑ Inventory monitoring with low-stock alerts.
- ❑ AI-based demand forecasting.
- ❑ Interactive dashboard visualisations.
- ❑ Automated report generation and export.
- ❑ AI-powered business recommendations.

Key Dashboard Metrics

- ❑ ✓ Revenue Trends
- ❑ ✓ Profit Analysis
- ❑ ✓ Inventory Status
- ❑ ✓ Customer Growth
- ❑ ✓ Sales Forecasting
- ❑ ✓ Product Performance Analysis

5. Objectives

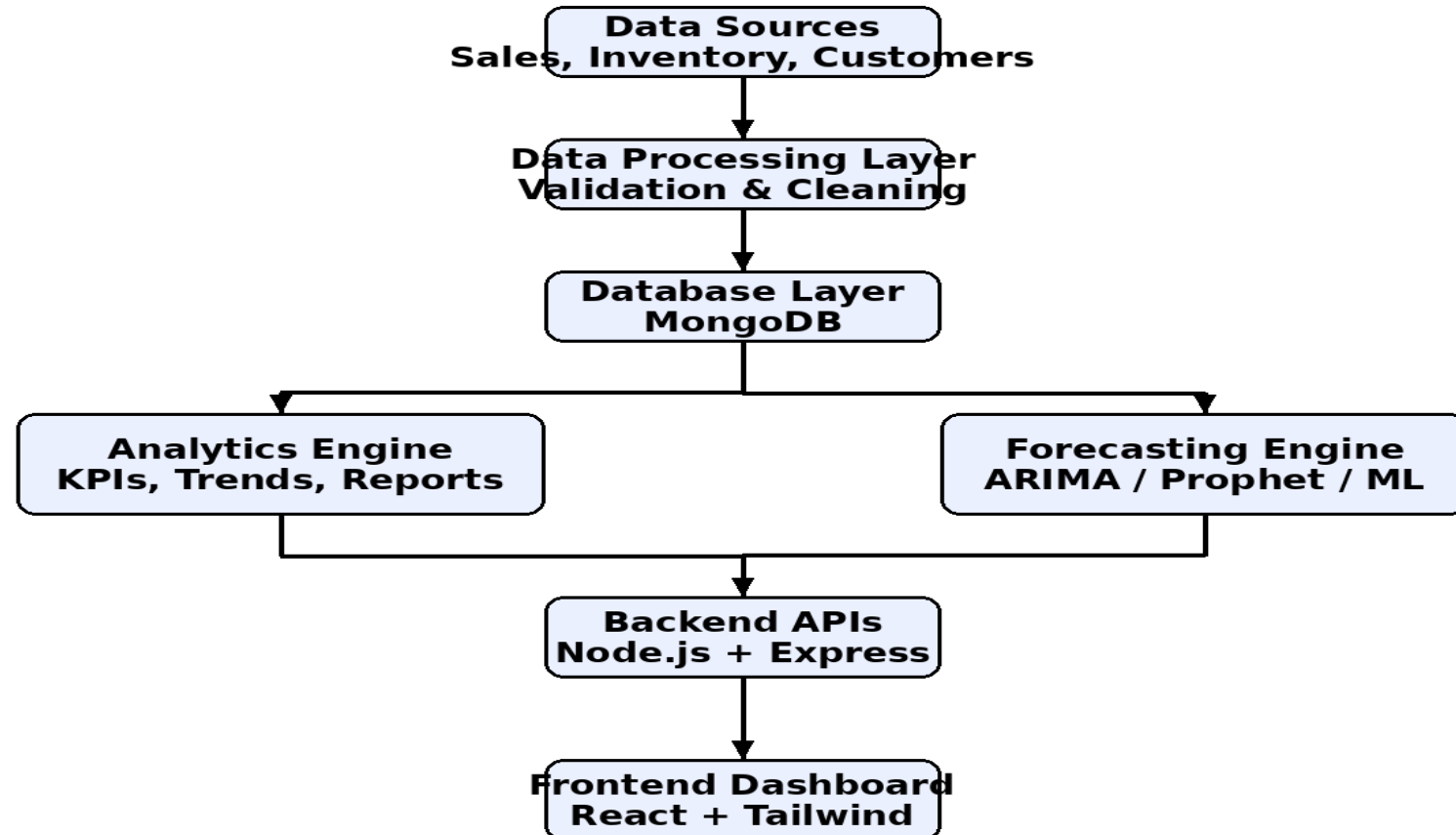
Project Objectives

- ❑ Eliminate manual reporting processes.
- ❑ Provide real-time business insights.
- ❑ Improve inventory monitoring and stock control.
- ❑ Generate accurate sales and revenue forecasts.
- ❑ Enable AI-driven recommendations.
- ❑ Support strategic planning through predictive analytics.
- ❑ Improve business decision-making using data visualisation.



6. System Architecture

Business Analytics Dashboard - System Architecture



7. Technology Stack

Frontend

- ✓ React.js
- ✓ Tailwind CSS
- ✓ JavaScript
- ✓ Recharts

Backend

- ✓ Node.js
- ✓ Express.js

Database

- ✓ MongoDB

Analytics & Forecasting

- ✓ Python
- ✓ Pandas
- ✓ NumPy
- ✓ Scikit-Learn
- ✓ ARIMA
- ✓ Prophet

Tools

- ✓ Git & GitHub
- ✓ VS Code
- ✓ Postman

8. Dashboard Modules

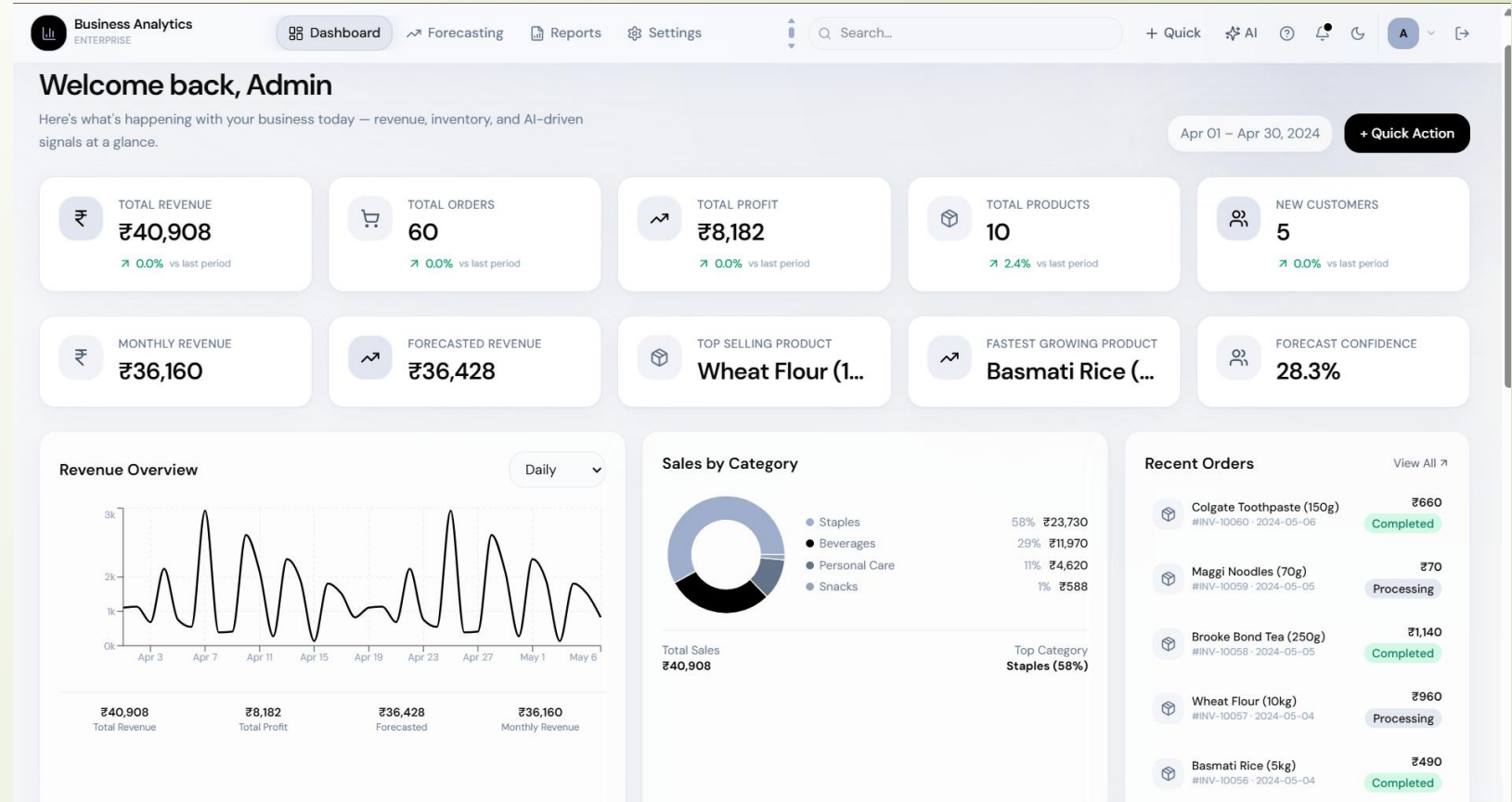
Dashboard

KPI Monitoring

Revenue Tracking

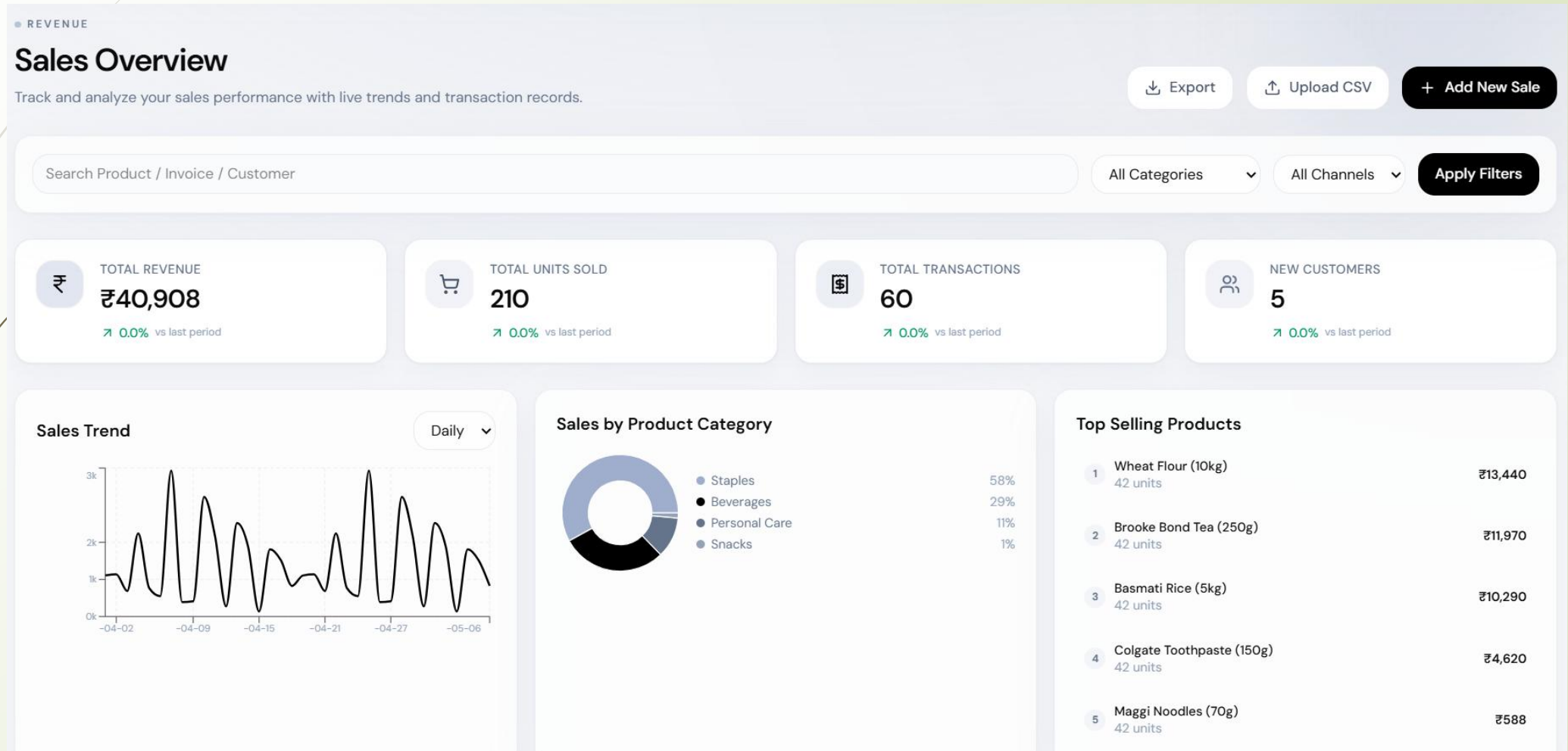
Category Analysis

Recent Order)



8.1 Sales Module

(Sales Records, Product Analytics, Customer Insights, Sales Trends)



8.2 Inventory Module

(Stock Management, Low Stock Alerts, Inventory Value Tracking)

STOCK

Inventory Overview

Track stock levels, manage products, and respond to low-stock alerts in real time.

Export

Import CSV

Add Product

TOTAL PRODUCTS
10

TOTAL STOCK (UNITS)
3,042

LOW STOCK ITEMS
1

OUT OF STOCK
1

INVENTORY VALUE
₹3,41,185

Search Product / SKU

All Categories

All Warehouses

Apply Filters

Inventory List

Product	SKU	Category	Warehouse	Stock	Threshold	Status	Stock Value	Actions
Basmati Rice (5kg)	GR-001	Staples	Main Warehouse	480	50	In Stock	₹1,17,600	
Wheat Flour (10kg)	GR-002	Staples	Main Warehouse	310	40	In Stock	₹99,200	
Toor Dal (1kg)	GR-003	Staples	Main Warehouse	0	30	Out of Stock	₹0	
Refined Oil (1L)	GR-004	Cooking	West Warehouse	22	25	Low Stock	₹3,410	
Sugar (1kg)	GR-005	Staples	Main Warehouse	540	60	In Stock	₹22,680	
Brooke Bond Tea (250g)	GR-006	Beverages	Main Warehouse	95	30	In Stock	₹27,075	

Stock Status Summary

In Stock

8 (80%)

Low Stock

1 (10%)

Out of Stock

1 (10%)

No Stock Data

0 (0%)

Low Stock Alerts

Toor Dal (1kg)

0 units left

Refined Oil (1L)

22 units left

[View All](#)

Recent Stock Movements

Sugar (1kg)

8.3 Forecasting Module

(Demand Prediction, Revenue Forecasting, AI Recommendations)

Demand Forecasting

AI-powered demand predictions to optimize inventory and revenue planning.

Basmati Rice (5kg)

Linear Regression

Quarterly

18/06/2026

3 Months

Generate Forecast

Model Performance Metrics (Linear Regression)

MAPE

100.0%

Lower is better

RMSE

₹2

Lower is better

Accuracy

0.0%

Higher is better



FORECASTED SALES

₹90



AVG DAILY DEMAND

1.24 units



PEAK DEMAND DAY

Thursday



REVENUE FORECAST

₹17,532



RECOMMENDED STOCK

108 units

Demand Forecast

— Historical — Forecast



Forecast Summary

May 05 - May 11
7 units forecast

₹1,364
₹0 - ₹14

May 12 - May 18
7 units forecast

₹1,364
₹0 - ₹14

May 19 - May 25
7 units forecast

₹1,364
₹0 - ₹14

8.4 Reports Module

(CSV Export PDF Export Automated Reports)

Reports

Generate, export, and schedule business reports with advanced filters.

Export CSV

Export PDF

Sales Report

Last 7 Days

All Categories

Generate Report

₹

TOTAL REVENUE

₹40,908

0.0% vs last period

0

TOTAL ORDERS

60

0.0% vs last period

U

UNITS SOLD

210

0.0% vs last period

A

AVG ORDER VALUE

₹682

Report Preview — Sales Report

Last 7 Days

Date	Product	Category	Orders	Units Sold	Revenue	Profit
2024-04-01	Basmati Rice (5kg)	Staples	1	1	₹245	₹49
2024-04-01	Wheat Flour (10kg)	Staples	1	2	₹640	₹128
2024-04-02	Brooke Bond Tea (250g)	Beverages	1	3	₹855	₹171
2024-04-02	Maggi Noodles (70g)	Snacks	1	4	₹56	₹11
2024-04-03	Colgate Toothpaste (150g)	Personal Care	1	5	₹550	₹110
2024-04-04	Basmati Rice (5kg)	Staples	1	6	₹1,470	₹294
2024-04-04	Wheat Flour (10kg)	Staples	1	1	₹320	₹64

Email Report

Enter email address

Send Report

Schedule Report

Weekly

Recipient email

Schedule Report

8.5 AI Insights Module

● INTELLIGENCE

AI Insights

Predictive recommendations powered by your live business data — optimize inventory, revenue, and customer growth.

✦ Live analysis

**Recommendation Engine**
Updated from dashboard analytics pipeline

●

Stock Shortage: Brooke Bond Tea (250g)

Potential stock depletion in 40 days. Reorder recommended. Current stock: 95. Recommended reorder: 0 units.

↗

●

Opportunity: Basmati Rice (5kg)

Predicted demand expected to reach 30 units next month. Profit is high (₹2027). Growth velocity is strong.

⚡

●

Opportunity: Wheat Flour (10kg)

Predicted demand expected to reach 55 units next month. Profit is high (₹2647). Growth velocity is strong.

🎯

●

Revenue Outlook

Projected 30-day revenue forecast is ₹36,428 based on the selected ARIMA model.

💡

12 SIGNALS
Demand signals
Cross-category velocity and seasonality patterns.

3 ACTIVE
Risk alerts
Stockouts and margin compression watchlist.

5 ACTIONS
Growth levers
High-LTV segments and upsell opportunities.

9. AI Forecasting & Analytics

Forecast Outputs

- Forecasted Sales
- Revenue Forecast
- Peak Demand Day
- Recommended Stock Levels

Performance Metrics

- MAPE (Mean Absolute Percentage Error)
- RMSE (Root Mean Square Error)
- Forecast Accuracy

Business Benefits

- Reduce stockouts
- Improve inventory planning
- Increase revenue opportunities
- Optimise purchasing decisions



Results & Value Addition

Results Achieved

- ✓ Centralized Business Monitoring
- ✓ Automated Analytics
- ✓ Inventory Optimization
- ✓ Predictive Sales Forecasting
- ✓ Faster Report Generation

Value Added Features

- 🔥 AI-Based Forecasting
- 🔥 Inventory Replenishment Recommendations
- 🔥 Interactive KPI Dashboard
- 🔥 Automated Reporting
- 🔥 Data-Driven Decision Support
- 🔥 Revenue & Profit Trend Analysis
- 🔥 Customer & Product Performance Tracking



Conclusion & Future Scope

? The Business Analytics Dashboard successfully integrates Sales Analytics, Inventory Management, Reporting, and Predictive Forecasting into a single platform.

? The system enables organisations to monitor performance, forecast future demand, and make informed business decisions through AI-powered analytics.

? **Future Scope**

- ✓ Real-Time Data Streaming
- ✓ Role-Based Access Control
- ✓ Advanced ML Models (LSTM)
- ✓ Cloud Deployment (AWS/Azure)
- ✓ Customer Segmentation
- ✓ Anomaly Detection
- ✓ Mobile Application Support
- ✓ AI Chat Assistant for Business Insights



GitHub Link

❓ <https://github.com/Rayyan-sheikh2748/Business-Analytics-#>

Thank You!

- ❓ Any Questions?
- ❓ **Business Analytics Dashboard**
(Sales & Inventory Management with Predictive Insights)